

From Following Patterns to Writing Them: Learning as Model Growth

Summary

Every crocheter walks a path from following patterns step by step, to modifying patterns confidently, to designing original patterns from scratch. This progression is not just about accumulating skills — it is about building an increasingly complex and accurate **generative model**. Each project you complete, each mistake you learn from, each new technique you master adds depth to your internal model of how stitches create fabric and fabric creates objects. This module traces that learning journey through the lens of Active Inference.

Learning Objectives

By the end of this module, you will be able to:

1. Trace the progression from following patterns to writing them as increasing generative model complexity
 2. Explain how each completed project updates and enriches the crocheter's generative model
 3. Describe pattern design as the ultimate expression of generative model mastery
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Key Concepts

1. The Three Stages of Pattern Literacy

Stage 1 — Following: The beginner follows a pattern line by line, stitch by stitch. Their generative model is sparse — they can decode the current instruction but cannot predict what comes next or visualize the whole. Each stitch is a separate act of decoding and executing, with little sense of the larger structure.

Stage 2 — Modifying: The intermediate crocheter reads the pattern as a whole, understands its logic, and makes deliberate modifications. They change yarn, adjust sizing, substitute stitches, and know what will happen. Their generative model is rich enough to simulate the effects of changes before making them.

Stage 3 — Designing: The advanced crocheter can create original patterns — they generate the generative model itself. They think in terms of construction logic, shaping mathematics, and stitch properties to create new objects from imagination. Their model is so well-trained that it can run generatively: producing novel outputs that have never been made before.

This progression mirrors a fundamental principle of Active Inference: **learning is model growth**. The beginner's model is shallow and narrow. Each project adds depth, breadth, and precision until the model can not only decode external patterns but generate new ones.

2. How Projects Update Your Model

Each completed project is a learning event. Specifically:

Successful projects confirm and reinforce predictions: - "Worsted weight on a 5mm hook at my tension produces this gauge" — confirmed for the 10th time, becoming a high-confidence prior - "Shell stitch creates a scalloped edge" — reinforced by consistent results

Failed projects update the model at precisely the points where it was wrong: - The sweater that does not fit teaches about the relationship between gauge and ease - The amigurumi that is floppy teaches about stuffing density and stitch tightness - The blanket that curls teaches about the difference between sc and dc fabric behavior

Novel projects extend the model into new territory: - Your first colorwork project teaches about carrying yarn, tension changes at color transitions, and managing multiple balls - Your first garment teaches about construction, shaping, and the relationship between flat pieces and three-dimensional form - Your first lace project teaches about how open stitches create pattern through negative space

In Active Inference, all of these are forms of **Bayesian model updating**: the posterior after one project becomes the prior for the next. Over time, your model accumulates a library of

well-calibrated predictions about yarn behavior, stitch properties, construction methods, and finishing techniques.

3. Error-Driven Learning: Why Mistakes Are the Best Teacher

There is a saying in crochet circles: "If you have never had to frog, you have never learned anything." From an Active Inference perspective, this is precisely right.

Prediction errors — the moments when your model's predictions do not match reality — are the primary driver of learning. They signal exactly where your model needs updating:

- A missed increase teaches you to count more carefully (updating a perceptual habit)
- A gauge that drifted teaches you about tension consistency (updating a motor model)
- A pattern that produced a different shape than expected teaches you about construction logic (updating a structural model)

Errors that are small and frequent (wrong stitch count, needing to tink back a few stitches) provide fine-grained model updates. Errors that are large and rare (a completely unwearable garment) provide dramatic model revisions. Both are essential to learning.

The emotional sting of frogging a large section is, in a sense, the felt experience of a large prediction error. It hurts because the model was confident in its predictions and was proven wrong. This emotional signal ensures the model update is deep and lasting.

4. From Consumer to Creator: Pattern Design as Model Mastery

Writing a crochet pattern is the ultimate test of your generative model. To design a pattern, you must:

1. **Generate:** Conceive of a finished object that does not yet exist — your model must be able to produce novel predictions
2. **Decompose:** Break the object into construction steps — working backward from the finished form to the first stitch
3. **Calculate:** Determine stitch counts, shaping schedules, and material requirements — running your model's computational capabilities

4. **Encode:** Write instructions that another crocheter can follow — translating your internal model into external symbols
5. **Test:** Make the object and verify that your predictions were correct — the ultimate model validation
6. **Refine:** Edit the pattern based on test results and feedback — iterative model improvement

Pattern design requires the model to operate in the **generative direction** — producing new predictions rather than just consuming and verifying existing ones. This is why design feels fundamentally different from following: you are running the model forward into uncharted territory.

5. Community Learning: How Crochet Circles Accelerate Model Growth

Learning does not happen in isolation. Crochet circles, workshops, and online communities accelerate model growth through several mechanisms:

- **Observation:** Watching another crocheter work lets you observe their model's outputs (their technique, speed, problem-solving) and incorporate insights into your own model
 - **Instruction:** Direct teaching transmits model components that would take much longer to learn through trial and error
 - **Feedback:** Having someone else look at your work provides external observations that may detect prediction errors your own model missed
 - **Shared projects:** Crochet-alongs and mystery KALs/CALs provide structured learning experiences that build specific model capabilities
 - **Pattern exchange:** Sharing patterns is literally sharing generative models, allowing each crocheter to benefit from others' model development
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Applications: Accelerating Your Learning

The Learning Timeline

Map your own crochet learning journey. What was your first project? When did you first modify a pattern? When did you first design something original? For each milestone, identify what capability your generative model gained.

The Deliberate Practice Plan

Identify the weakest area of your generative model (gauge prediction? construction logic? shaping math?) and design a series of small projects specifically to strengthen that area. This is targeted model updating.

The Teaching Experiment

Teach a basic stitch or technique to someone who has never crocheted. Notice what you have to make explicit — every explanation reveals a piece of your model that you had internalized so deeply you forgot it was learned.

Conclusion

Learning in crochet is the progressive enrichment of your generative model. From the sparse, stitch-by-stitch model of the beginner to the rich, generative model of the designer, every project, every mistake, and every new technique adds to your capacity to predict, create, and adapt. Understanding this process helps you learn more efficiently — by seeking out the errors that drive model growth and the novel experiences that extend your model into new territory.

In the next module, we explore communication — how crocheters share their generative models through notation systems, pattern conventions, and community platforms.

Key Terms

Term	Crochet Meaning	Active Inference Meaning
Following a pattern	Working step by step from instructions	Consuming an external generative model
Modifying a pattern	Making deliberate changes	Updating model parameters while preserving structure
Designing a pattern	Creating original instructions	Generating a new generative model
Frogging a failure	Undoing work that went wrong	Processing a large prediction error
Practice	Repeating techniques to build skill	Training the generative model through experience
Teaching	Showing someone how to crochet	Transmitting model components to another agent